

Curriculum Vitae

Personal Data

Title	Prof. Dr.-Ing. (habil.)
First Name	Christoph
Name	Benzmüller
Current position	Full professor (Univ.Prof, chair) at University of Bamberg
Current institutions/ sites, country	Since 02/22: University (U) of Bamberg (GER). 10/23-03/24: Visiting/Guest Prof., BITS Pilani Dubai (UAE). Since 08/21: Adjunct (apl.) Prof., FU Berlin (GER). Since 10/20: Consultant/Scientific Advisor, Freelancer (INTL.)
Identifiers/ORCID	0000-0002-3392-3093

Qualifications and Career

Stages	Periods and Details
Degree programme	10/89–04/95: Dipl.-Inform., Computer Science, Saarland U, Germany
Doctorate	04/1995–07/1999: Saarland U (GER). Dissertation: "Equality and Extensionality in Higher-Order Theorem Proving". Supervisors: Prof. J. Siekmann, Prof. M. Kohlhase (now FAU). Mentors: Prof. P. Andrews, Prof. F.Pfenning (both CMU, USA).
Stages of academic/ professional career	2020–2020: First UNA Europa Chair, Maths, AI & CS, FU Berlin (GER) 2018–2020: Guest Prof., Maths, AI & CS, FU Berlin (GER) 2017–2019: Visiting Scholar, AI & CS, U Luxembourg (LUX) 2015–2016: Visiting Scholar, Philosophy, Stanford U (USA) 2012–2017: DFG Heisenberg Fellow, FU Berlin (GER) Since 02/2012: Venia Legendi in Maths & CS, FU Berlin (GER) 2011–2011: DFG Return Scholarship, FU Berlin (GER) 2008–2009: Full Prof., AI & CS, Intl. U in Germany, Bruchsal (GER) 2006–2007: Senior Research Fellow, AI & CS, U Cambridge, (GBR) 2001-2008: Adjunct Prof./Hochschuldozent (C1 & C2), Saarland U (GER) 2000–2000: PostDoc, AI & CS, U Birmingham & U Edinburgh, (GBR) 1996–1997: Short-term Scholar, Maths, CMU (USA)

Engagement in the Research System

- High-end Foreign Expert in 2023-24 (for AI, Logic & Ethics) at Zhejiang University, China
- Course representative & implementation: New BSc programme AI & Data Science at U Bamberg
- Member of several professorship hiring committees at different universities
- Chair, organiser, PC member of numerous intl. conferences and workshops (e.g. IJCAR, CADE)
- Editor and editorial board member of different intl. journals (e.g. LJ.IGPL, MCS, FLAP, Historiae Logicae)
- Member of various natl./intl. research organisations, including Federation of German Scientist (VDW)
- National Contact Point in Germany of the CLAIRE-Network
- Vice-president 2015-2021 of CADE Inc. (Conference of Automated Deduction)

Supervision of Researchers in Early Career Phases

- Dr. Tarek Besold: Mentor of habilitation procedure/committee at U Bamberg
- Dr. Dr. Andrea Vestrucchi: awarded as guest professor in my group at U Bamberg

- Dr. Martin Aleksandrov: awarded with DFG own position (Eigene Stelle) in my group at FU Berlin
- Dr. Daniel Kirchner: lead position at Ethereum Foundation and PostDoc in my group at U Bamberg
- Dr. Alexander Steen: awarded with GI-Junior-Fellowship in 2018, now Jun.Prof. at U Greifswald

Completed Doctorates (selection, last 5 years)

- 2015-2018: Alexander Steen, Extens. Paramodulation for Higher-Order Logic and its Effective Implementation Leo-III (s.c.I.; afterwards: GI Junior Fellow, PostDoc U Luxemb., Jun.Prof. U Greifswald)
- 2018-2022: Daniel Kirchner, Computer-Verified Foundations of Metaphysics and an Ontology of Natural Numbers in Isabelle/HOL (s.c.I.; afterwards: Ethereum Foundation & PostDoc U Bamberg)
- 2016-2020: Ali Farjami, Discursive Input/Output Logic: Deontic Modals, and Computation (U Luxembourg, co-supervisor; afterwards: PostDoc U Luxembourg)
- I was supervisor, co-supervisor, committee member, mentor of ≥ 25 PhD and habilitation projects

Ongoing Doctorates

- since 1999: David Fuenmayor, Universal Reasoning/Comp. Hermeneutics (≥ 30 papers: [RG](#), [DBLP](#))
- since 2020: Rosae Martin Pena, Ethical Decision-Making in Trustworthy Human-AI Interaction: Linking Counterfactual Explanations and Regret (thesis already submitted; now PostDoc at DFKI Berlin)
- since 2020: Veronika Solopova, Hybrid AI Systems in Automated Content Moderation and Analysis (thesis will be submitted in winter; now already on PostDoc position at TU Berlin)
- I am co-supervisor or mentor of various further doctorates at different universities

Scientific Results

Category A (peer-reviewed, chronological order)

1. **Benzmüller, C.**, Fuenmayor, D., Steen, A. & Sutcliffe, G. Who Finds the Short Proof? *Logic Journal of the IGPL*. doi:[10.1093/jigpal/jzac082](#) (2023)
2. Steen, A., Sutcliffe, G., Scholl, T. & **Benzmüller, C.** *Solving Modal Logic Problems by Translation to Higher-order Logic* in *CLAR 2023* (Springer, 2023). doi:[10.1007/978-3-031-40875-5_3](#). Best Paper Award
3. **Benzmüller, C.** Symbolic AI and Gödel's Ontological Argument. *Zygon*. doi:[10.1111/zygo.12830](#) (2022)
4. **Benzmüller, C.** & Reiche, S. Automating Public Announcement Logic with Relativized Common Knowledge as a Fragment of HOL in LogiKey. *Journal of Logic and Computation*. doi:[10.1093/logcom/exac029](#) (2022)
5. Steen, A. & **Benzmüller, C.** Extensional Higher-Order Paramodulation in Leo-III. *Journal of Automated Reasoning*. doi:[10.1007/s10817-021-09588-x](#) (2021)
6. **Benzmüller, C.**, Parent, X. & van der Torre, L. Designing Normative Theories for Ethical and Legal Reasoning: LogiKey Framework, Methodology, and Tool Support. *Artificial Intelligence*. doi:[10.1016/j.artint.2020.103348](#) (2020)
7. **Benzmüller, C.** Universal (Meta-)Logical Reasoning: Recent Successes. *Science of Computer Programming*. doi:[10.1016/j.scico.2018.10.008](#) (2019)
8. **Benzmüller, C.**, Farjami, A. & Parent, X. *A Dyadic Deontic Logic in HOL* in *DEON 2018* (College Publications, 2018). John-Jules Meyer Best Paper Award
9. **Benzmüller, C.**, **Brown, C.** & Kohlhase, M. Cut-Simulation and Impredicativity. *Logical Methods in Computer Science*. doi:[10.2168/LMCS-5\(1:6\)2009](#) (2009)
10. **Benzmüller, C.**, **Brown, C.** & Kohlhase, M. Higher-Order Semantics and Extensionality. *Journal of Symbolic Logic*. doi:[10.2178/jsl/1102022211](#) (2004)

Category B (others, chronological order)

1. **Benzmüller, C.** *et al.* LogiKEy Workbench: Deontic Logics, Logic Combinations and Expressive Ethical and Legal Reasoning (Isabelle/HOL Dataset). *Data in Brief* **33**, 1–10. doi:[10.1016/j.dib.2020.106409](https://doi.org/10.1016/j.dib.2020.106409) (2020)
2. Fuenmayor, D. & **Benzmüller, C.** Formalisation and Evaluation of Alan Gewirth's Proof for the Principle of Generic Consistency in Isabelle/HOL. *Archive of Formal Proofs*. Note: data publication, 1–15 (2018)
3. **Benzmüller, C.**, Parent, X. & Ricca, F. Report on the Second International Joint Conference on Rules and Reasoning. *AI Magazine* **40**, 73–74. doi:[10.1609/aimag.v40i2.2888](https://doi.org/10.1609/aimag.v40i2.2888) (2019)

Academic Distinctions

Awards (selection)

- High-end Foreign Expert in 2023-24 (for AI, Logic & Ethics) at Zheijang University, China
- Best paper awards, e.g. at DEON 2018 and CLAR 2023 conferences
- World champion in Higher-order Automated Theorem Proving (LTB) in 2019 with LEO-III prover
- FU Berlin's central teaching award for the lecture concept on *Computational Metaphysics*, 2016
- Shortlisted in 2013 for *Amalia Preis für Neues Denken*, Weimar, Germany
- DFG Heisenberg fellowship of the DFG, 2012–2017
- World champion in Higher-order Automated Theorem Proving in 2010 with LEO-II prover
- PhD scholarship from 1996 to 1998 of the Studienstiftung des Deutschen Volkes

Invited talks & tutorials (selection of some recent keynotes)

- Keynote: Reasonable, Trusted AI through Symbolic Ethico-legal Control and Reflection? Florida Artificial Intelligence Research Society (FLAIRS-36), Clearwater Beach, Florida, USA, 2023.
- Keynote: Reasonable, trusted AI requires hybrid techniques. International Conference on Computational Intelligence and Network Systems (CINS 2023), Dubai, UAE, 2023.
- Keynote: Bridging Legal Argumentation and Automated Theorem Proving. Lawgorithm: AI, Ethics and Law Seminar, University of Sao Paulo, 2022.
- Keynote: Ethico-legal governance of intelligent artificial agents – Can post-hoc normative reasoning competencies prevent AI systems from going rogue? Plenary at ZJULogAI, Hangzhou, China, 2020.
- Panelist: Una Europa Talk, Resistance is futile. You will be digitalized! Berlin, Helsinki, Leuven, 2020.
- Keynote: Computational Metaphysics: New Insights on Gödel's Ontological Argument and Modal Collapse. Formal Methods and Science in Philosophy III, Dubrovnik, Croatia, 2019.
- Keynote: Human vs. Nonhuman – The Need for Ethical Intelligent Systems. Mecila & Goethe-Institut São Paulo, Brazil, 2019.
- Keynote: Experiments in Universal Logical Reasoning, 16th International Workshop on Satisfiability Modulo Theories (SMT 2018), Oxford, UK, July 2018.
- Keynote: A Flexible Infrastructure for Normative Reasoning, 14th International Conference (DEON 2018), Utrecht, The Netherlands, 2018.
- Keynote: What has the Mechanisation of Category Theory in Common with Proving God's Existence? 3. BMG Tag, Berliner Mathematische Gesellschaft e.V., Berlin, Germany 2017.
- Overall: ≥ 18 keynotes at conferences & workshops; ≥ 100 invited presentations & tutorials

Other Information

Some of my research has attracted enormous media interest, for example in TV productions such as TerraX (4 million viewers), Spiegel (online) magazine, newspapers such as Die Welt, FAZ, Tagesspiegel, etc. I have won several science slams, including the [FU Berlin Science Slam](#) as part of Wissensstadt Berlin event in 2021 and the [Science Slam of the U Bamberg](#) in 2022; however, my probably best science slam was in 2019: [#71 Science Slam Berlin](#) at the famous club “SO36” in Berlin-Kreuzberg.